

Understanding Compound Interest

Albert Einstein is thought to have said, “Compound interest is the eighth wonder of the world. He who understands it *earns* it. He who doesn’t *pays* it. Compound interest is the most powerful force in the universe.” What makes compound interest so unbelievably powerful? Time. Investing early and consistently over a long period of time allows you to take advantage of the two most powerful forces in all of finance: time and compound interest. When you put money into an investment and let it sit awhile, your money grows from the interest you earn. If you put \$1,000 in originally, you might have \$1,100 a year later. That extra \$100 is the result of the interest you earned. What happens in the second year? Well, now you aren’t earning interest on \$1,000; you’re earning interest on \$1,100. So the money that grew from interest in the first year is now earning interest itself. That’s what *compounding* means—it’s when your interest earns interest. Your money is basically creating *more* money, and it keeps doing that on its own—morning, noon, and night.

The effect that time and compound interest have on your money is mind-blowing. Let me put it into perspective for you. What if I offered you a choice: you can have \$1 million cash *right now* and walk away, or you can have one penny right now. If you choose the penny, you can come back tomorrow, and I’ll double it, leaving you with two pennies. Then I’ll double it again the next day, bringing your grand total to four cents after three days. I’ll keep doubling it every day for a month—thirty-one days in all. Which of these two choices sounds like the better deal? Most people would take the \$1 million and run away laughing. Who would choose pennies over a cool million? Here’s the thing, though: those pennies keep on doubling. After a week, you’d have \$1.27, which doesn’t sound great. After two weeks, you’d be up to \$163.83. After three weeks, things would start to get interesting: your \$163.83 would have jumped up to \$20,971.51. One week later, at day twenty-eight, you’d be up to \$2,684,354.55. You’re a